bst-map

EECS 280 Project 6: Binary Search Trees and Maps

Winter 2026 release.

Project due 8:00pm EST Tuesday April 21, 2026.

You may work alone or with a partner (partnership guidelines).

Introduction

In this project, you implement a map container (similar to `std::map`) based on an underlying binary search tree data structure. The learning goals of this project include functors, templates, recursion, binary search trees, and associative containers.

Setup

Set up your visual debugger and version control, then submit to the autograder.

Visual debugger

During setup, name your project `p6-bst-map`. Use this starter files link: `https://eecs280staff.github.io/bst-map/starter-files.tar.gz`

VS Code	Visual Studio	Xcode

You should end up with a folder with starter files that look like this. You may have already renamed files like `Map.hpp.starter` to `Map.hpp`. You may also have a `main.cpp` file after following the setup tutorial. If so, delete the file.

```
$ ls
BinarySearchTree.hpp.starter        Map_compile_check.cpp
BinarySearchTree_compile_check.cpp  Map_public_tests.cpp
BinarySearchTree_public_tests.cpp   Map_tests.cpp.starter
BinarySearchTree_tests.cpp.starter  TreePrint.hpp
Makefile                            unit_test_framework.hpp
```

```
Map.hpp.starter
```

Here's a short description of each starter file.

File(s)	Description
`BinarySearchTree.hpp.starter`	Starter code for `BinarySearchTree`.
`BinarySearchTree_tests.cpp.starter`	Your `BinarySearchTree` unit tests.
`BinarySearchTree_public_tests.cpp`	A small test for `BinarySearchTree`
`BinarySearchTree_compile_check.cpp`	Compile check test for `BinarySearchTree`
`TreePrint.hpp`	Test helper function for printing trees.
`Map.hpp.starter`	Starter code for `Map`.
`Map_tests.cpp.starter`	Your `Map` unit tests.
`Map_public_tests.cpp`	Your `Map` unit tests.
`Map_compile_check.cpp`	Compile check test for `Map`.
`Makefile`	Helper commands for building.
`unit_test_framework.hpp`	A simple unit-testing framework.

Version control

Set up version control using the Version control tutorial.

After you're done, you should have a local repository with a "clean" status and your local repository should be connected to a remote GitHub repository.

```
$ git status
On branch main
Your branch is up-to-date with 'origin/main'.

nothing to commit, working tree clean
$ git remote -v
origin  https://github.com/awdeorio/p6-bst-map.git (fetch)
origin  https://githubcom/awdeorio/p6-bst-map.git (push)
```

You should have a `.gitignore` file (instructions).

```
$ head .gitignore
# This is a sample .gitignore file that's useful for C++ projects.
```

```
...
```

Group registration

Register your partnership (or working alone) on the Autograder. Then, submit the code you have.

BinarySearchTree

A binary search tree supports efficiently storing and searching for elements.

Write implementations in `BinarySearchTree.hpp` for each `_impl` function. The file already contains function stubs and you should replace the `assert(false)` with your code. For example:

BinarySearchTree.hpp

```
static bool empty_impl(const Node *node) {
  assert(false);  // Replace with your code
}
```

Run the public Binary Search Tree tests.

```
$ make BinarySearchTree_compile_check.exe
$ make BinarySearchTree_public_tests.exe
$ ./BinarySearchTree_public_tests.exe
```

Write tests for `BinarySearchTree` in `BinarySearchTree_tests.cpp` using the Unit Test Framework. You'll submit these tests to the autograder. See the Unit Test Grading section.

```
$ make BinarySearchTree_tests.exe
$ ./BinarySearchTree_tests.exe
```

Submit `BinarySearchTree.hpp` and `BinarySearchTree_tests.cpp` to the autograder.

Setup

Rename these files (VS Code (macOS), VS Code (Windows), Visual Studio, Xcode, CLI):

- `BinarySearchTree.hpp.starter` -> `BinarySearchTree.hpp`
- `BinarySearchTree_tests.cpp.starter` -> `BinarySearchTree_tests.cpp`

The BinarySearchTree tests should compile and run. The public tests and compile check will fail until you implement the functions. The test you write (`BinarySearchTree_tests.cpp`) will pass because the starter file only contains `ASSERT_TRUE(true)` .

```
$ make BinarySearchTree_compile_check.exe
```

```
$ make BinarySearchTree_public_tests.exe
$ ./BinarySearchTree_public_tests.exe
$ make BinarySearchTree_tests.exe
$ ./BinarySearchTree_tests.exe
```

Configure your IDE to debug either the public tests or your own tests.

	Public tests	
VS Code (macOS)	Set program name to: `${workspaceFolder}/BinarySearchTree_public_tests.exe`	Set program nam `${workspaceFold`
VS Code (Windows)	Set program name to: `${workspaceFolder}/BinarySearchTree_public_tests.exe`	Set program nam `${workspaceFold`
Xcode	Include compile sources: `BinarySearchTree_public_tests.cpp`	Include compile s `BinarySearchTre`
Visual Studio	Exclude files from the build: • Include `BinarySearchTree_public_tests.cpp` • Exclude any other tests	Exclude files from • Include `Bina` • Exclude any

Template Parameters

`BinarySearchTree` has two template parameters:

- `T` - The type of elements stored within the tree.
- `Compare` - The type of comparator object (a functor) that should be used to determine whether one element is less than another. The default type is `std::less<T>`, which compares two `T` objects with the `<` operator. To compare elements in a different fashion, a custom comparator type must be specified.

No Duplicates Invariant

In the context of this project, duplicate values are NOT allowed in a BST. This does not need to be the case, but it avoids some distracting complications.

Sorting Invariant

A binary search tree is special in that the structure of the tree corresponds to a sorted ordering of elements and allows efficient searches (i.e. in logarithmic time).

Every node in a well-formed binary search tree must obey this sorting invariant:

- It represents an empty tree (i.e. a null `Node*`).

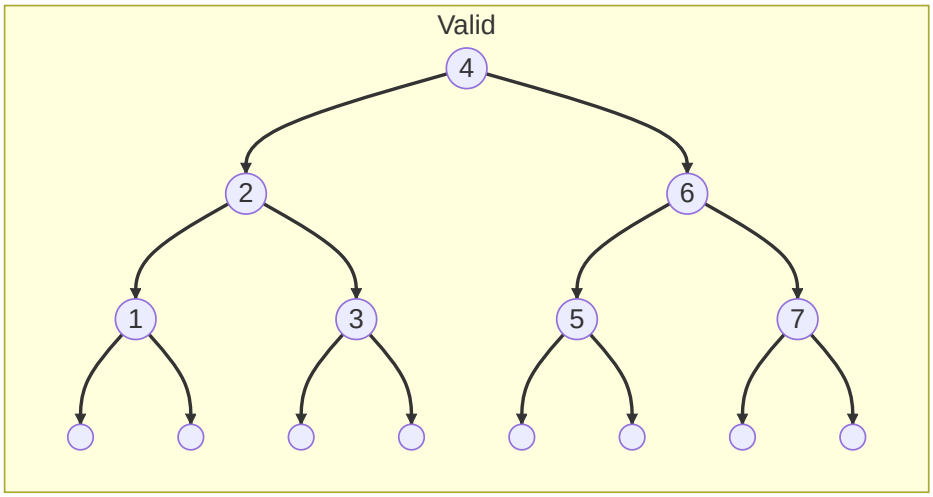
- OR -

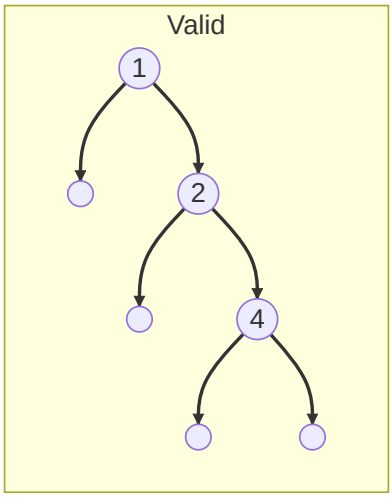
- The left subtree obeys the sorting invariant, and every element in the left subtree is less than the root element (i.e. this node).

  - AND -

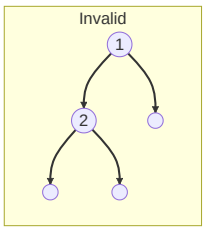
  The right subtree obeys the sorting invariant, and the root element (i.e. this node) is less than every element in the right subtree.

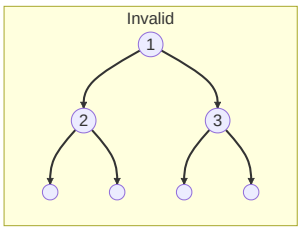
Put briefly, go left and you'll find smaller elements. Go right and you'll find bigger ones. For example, the following are all well-formed sorted binary trees:

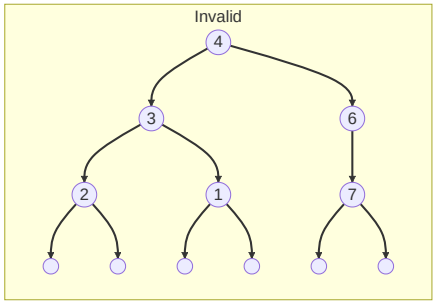


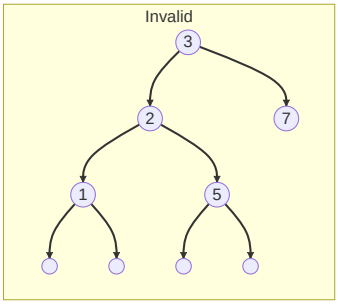


While the following are not:









ProTip: When writing tests for `check_sorting_invariant()`, you can use an iterator to break the invariant. For example:

```
BinarySearchTree<int> b;
```

```
b.insert(1);
b.insert(0);
// change first datum to 2, resulting in the first broken tree above
*b.begin() = 2;
ASSERT_FALSE(b.check_sorting_invariant());
```

Data Representation

The data representation for `BinarySearchTree` is a tree-like structure of nodes similar to that described in lecture. Each `Node` contains an element and pointers to left and right subtrees. The structure is self-similar. A null pointer indicates an empty tree. **You must use this data representation. Do not add member variables to `BinarySearchTree` or `Node`.**

Public Member Functions and Iterator Interface

The public member functions and iterator interface for `BinarySearchTree` are already implemented in the starter code. DO NOT modify the code for any of these functions. They delegate the work to private, static implementation functions, which you will write.

Implementation Functions

The core of the implementation for `BinarySearchTree` is a collection of private, static member functions that operate on tree-like structures of nodes. **You are responsible for writing the implementation of several of these functions.**

To disambiguate these implementation functions from the public interface functions, we have used names ending with `_impl`. (This is not strictly necessary, because the compiler can differentiate them based on the `Node*` parameter.)

There are a few keys to thinking about the implementation of these functions:

- **The functions have no idea that such a thing as the `BinarySearchTree` class exists**, and they shouldn't. A "tree" is not a class, but simply a tree-shaped structure of `Node`s. The parameter node points to the root of these nodes.
- A recursive implementation depends on the idea of **similar subproblems**, so a "subtree" is just as much a tree as the "whole tree". That means you shouldn't need to think about "where you came from" in your implementation.
- Every function should have a **base case**! Start by writing this part.
- You only need to think about one "level" of recursion at a time. Avoid thinking about the contents of subtrees and **take the recursive leap of faith**.

We've structured the starter code so that the first bullet point above is actually enforced by the language. Because they are `static` member functions, they do not have access to a receiver object (i.e. there's no `this` pointer). That means it's actually impossible for these functions to try to do something bad with the `BinarySearchTree` object (e.g. trying to access the `root` member variable).

Instead, the implementation functions are called from the regular member functions to perform specific operations on the underlying nodes and tree structure, and are passed only a pointer to the root `Node` of the tree/subtree they should work with.

The `empty_impl` function must run in constant time. It must must be able to determine and return its result immediately, without using either iteration or recursion. The rest of the implementation functions must be recursive. There are additional requirements on the kind of recursion that must be used for some functions. See comments in the starter code for details. **Iteration (i.e. using loops) is not allowed in any of the `_impl` functions**.

Using the Comparator

The `_impl` functions that need to compare data take in a comparator parameter called `less`. Make sure to use `less` rather than the `<` operator to compare elements!

The `insert_impl` Function

The key to properly maintaining the sorting invariant lies in the implementation of the `insert_impl` function - this is essentially where the tree is built, and this function will make or break the whole ADT. Your `insert_impl` function should follow this procedure:

1. Handle an originally empty tree as a special case.
2. Insert the element into the appropriate place in the tree, keeping in mind the sorting invariant. You'll need to compare elements for this, and to do so make sure to use the `less` comparator passed in as a parameter.
3. Use the recursive leap of faith and call `insert_impl` itself on the left or right subtree. Hint: You **do** need to use the return value of the recursive call. (Why?)

> ⚠ **Important:** When recursively inserting an item into the left or right subtree, be sure to replace the old left or right pointer of the current node with the result from the recursive call. This is essential, because in some cases the old tree structure (i.e. the nodes pointed to by the old left or right pointer) is not reused. Specifically, if the subtree is empty, the only way to get the current node to "know" about the newly allocated node is to use the pointer returned from the recursive call.

Technicality: In some cases, the tree structure may become unbalanced (i.e. too many nodes on one side of the tree, causing it to be much deeper than necessary) and prevent efficient operation for large trees. You don't have to worry about this.

Testing

Pro-tip: When writing tests for functions that return a `size_t` (which is an unsigned integer type), compare against an unsigned literal. For example:

```
BinarySearchTree<int> b;
ASSERT_EQUAL(b.height(), 0u);
```

Map

Write a map abstract data type (ADT). Map is an *associative container*, and works just like `std::map`.

Write implementations at the end of `Map.hpp` for the functions declared at the beginning of `Map.hpp`. The most important functions are `find`, `insert`, and the `[]` operator.

Your implementations should not require much code. Reuse the functionality provided by `BinarySearchTree`.

Run the public Map tests.

```
$ make Map_compile_check.exe
$ make Map_public_tests.exe
$ ./Map_public_tests.exe
```

Write tests for `Map` in `Map_tests.cpp` using the Unit Test Framework. While you should write your own tests for `Map` to ensure that your implementation is correct, you do not have to submit your tests to the autograder.

```
$ make Map_tests.exe
$ ./Map_tests.exe
```

Submit `Map.hpp` to the autograder. Don't forget to include the code you finished earlier, `BinarySearchTree.hpp` and `BinarySearchTree_tests.cpp`.

Setup

Rename these files (VS Code (macOS), VS Code (Windows), Visual Studio, Xcode, CLI):

- `Map.hpp.starter` -> `Map.hpp`
- `Map_tests.cpp.starter` -> `Map_tests.cpp`

Edit `Map.hpp`, adding a function stub for every function prototype in `Map`. Here are a few examples to get you started. We're using `K`, `V`, and `C` as shorthands for `Key_type`, `Value_type`, and `Key_compare`.

```
template <typename K, typename V, typename C>
bool Map<K, V, C>::empty() const {
  assert(false);
}

template <typename K, typename V, typename C>
typename Map<K, V, C>::Iterator Map<K, V, C>::find(const K& k) const {
  assert(false);
}

template <typename K, typename V, typename C>
V& Map<K, V, C>::operator[](const K& k) {
  assert(false);
}

template <typename K, typename V, typename C>
std::pair<typename Map<K, V, C>::Iterator, bool> Map<K, V, C>::insert(const Pair_type &val) {
  assert(false);
}
```

Now you should be able to compile and run the Map unit tests. The public tests will fail until you implement the functions.

```
$ make Map_compile_check.exe
$ make Map_public_tests.exe
$ ./Map_public_tests.exe
```

Configure your IDE to debug either the public tests or your own tests. Writing your own tests for Map is optional.

	Public tests	Your own tests
VS Code (macOS)	Set program name to: `${workspaceFolder}/Map_public_tests.exe`	Set program name to: `${workspaceFolder}/Map_tests`

	Public tests	Your own tests
VS Code (Windows)	Set program name to: `${workspaceFolder}/Map_public_tests.exe`	Set program name to: `${workspaceFolder}/Map_tests`
Xcode	Include compile sources: `Map_public_tests.cpp`	Include compile sources: `Map_tests.cpp`
Visual Studio	Exclude files from the build: • Include `Map_public_tests.cpp` • Exclude any other tests	Exclude files from the build: • Include `Map_tests.cpp` • Exclude any other tests

Map Examples

A map is an *associative container*. It stores two types, key and value. Our map works just like `std::map`.

```
Map<string, double> words;
std::map<string, double> words;
```

One way to use a map is a lot like an array.

```
words["hello"] = 1;
```

Maps store a `std::pair` type, which “glues” one key to one value. The computer science term is Tuple, a fixed-size heterogeneous container.

```
pair<string, double> tuple;
tuple.first = "world";
tuple.second = 2;
words.insert(tuple);
```

Here’s a more compact way to insert a pair.

```
words.insert({"pi", 3.14159});
```

The range-for loop makes it easier to iterate over a map.

```
for (const auto &kv : words) {
  const auto &word = kv.first; //key
  auto number = kv.second; //value
  cout << word << " " << number << endl;
```

```
}
```

You can check if a key is in the map. The `find()` function returns an iterator.

```
auto found_it = words.find("pi");
if (found_it != words.end()) {
  const auto &word = (*found_it).first; //key
  auto number = (*found_it).second; //value
  cout << "found " << word << " " << number << endl;
}
```

When using the `[]` notation, an element not found is automatically created. If the value type of the map is numeric, it will always be `0` by default.

```
cout << "bleh: " << words["bleh"] << endl;
```

Building on the BST

The operation of a map is quite similar to that of a BST. The additional consideration for a map is that we want to store key-value pairs instead of single elements, but also have any comparisons (e.g. for searching) only depend on the key and be able to freely change the stored values without messing up the BST sorting invariant. We can employ the has-a pattern using a `BinarySearchTree` as the data representation for Map:

- BST template parameter: `T`

  Instantiate with: `Pair_type`

  We've provided a using declaration in the starter code for `Pair_type`:

  `using Pair_type = std::pair<Key_type, Value_type>;`

  `std::pair` is basically like a struct that stores two objects together. `Key_type` and `Value_type` are whatever template parameters were used to instantiate `Map`.

- BST template parameter: `Compare`

  Instantiate with: `PairComp`

  You'll need to define your own comparator by declaring a functor type called `PairComp` (or whatever you want to call it) in your `Map` class. The overloaded `()` operator should accept two objects of `Pair_type` and return whether the key of the LHS is less than the key of the RHS (according to `Key_compare`).

Finally, we can even reuse the iterators from the BST class, since the interface we want (based on `std::map`) calls for iterators to yield a key-value pair when dereferenced. Since the element type `T` of the BST is our `Pair_type`, BST iterators will yield pairs and will work just fine. We've provided this `using` declaration with the starter code to make `Map::Iterator` simply an alias for iterators from the corresponding BST:

```
using Iterator = typename BinarySearchTree<Pair_type, PairComp>::Iterator;
```

Submission and Grading

Submit these files to the autograder.

- `BinarySearchTree.hpp`
- `BinarySearchTree_tests.cpp`
- `Map.hpp`

You do not have to submit `Map_tests.cpp` to the autograder.

This project will be autograded for correctness, comprehensiveness of your test cases, and programming style. See the style checking tutorial for the criteria and how to check your style automatically on CAEN.

Testing

Run all the unit tests. This includes the public tests we provided and the unit tests that you wrote.

```
$ make test
```

Pro-tip: Run commands in parallel with `make -j`.

```
$ make -j4 test
```

Unit Test Grading

We will autograde your `BinarySearchTree` unit tests.

Your unit tests must use the unit test framework.

A test suite must complete less than 5 seconds and contain 50 or fewer `TEST()` items. One test suite is one `_tests.cpp` file.

To grade your unit tests, we use a set of intentionally buggy instructor solutions. You get points for catching the bugs.

1. We compile and run your unit tests with a **correct solution**.
   - Tests that pass are **valid**.
   - Tests that fail are **invalid**, they falsely report a bug.
2. We compile and run all of your **valid** tests against each **buggy solution**.
   - If any of your tests fail, you caught the bug.
   - You earn points for each bug that you catch.

Requirements and Restrictions

DO	DO NOT
Create any private helper functions you want.	Modify the `BinarySearchTree` or `Map` public interfaces
Use any part of the STL except for containers in your `BinarySearchTree` and `Map` implementations.	Use STL containers in your implementation of `BinarySearchTree` or `Map`.
Use recursion for the BST `_impl` functions.	Use iteration for the BST `_impl` functions.
Follow course style guidelines.	Use non-const static or global variables.
Check for undefined behavior using address sanitizer and other tools	“It runs fine on my machine!”

Acknowledgments

Andrew DeOrio and James Juett wrote the original project and specification. Amir Kamil contributed to code structure, style, and implementation details. This project was developed for EECS 280, Fall 2016 at the University of Michigan. The classifier portion was split into a separate project in Fall 2024.